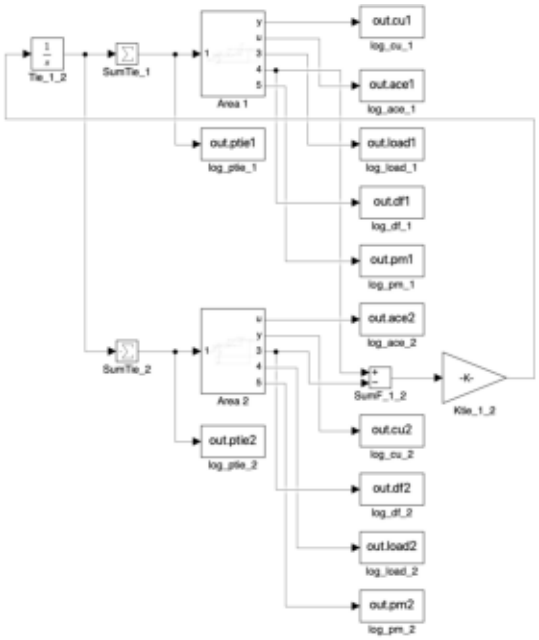
PID Load Frequency Control of a Two-Area Power System

Step load of 0.05 pu in Area 1 at t = 0 | linear state-space model and Simulink model



Simulink model (top level, two areas + tie-line)

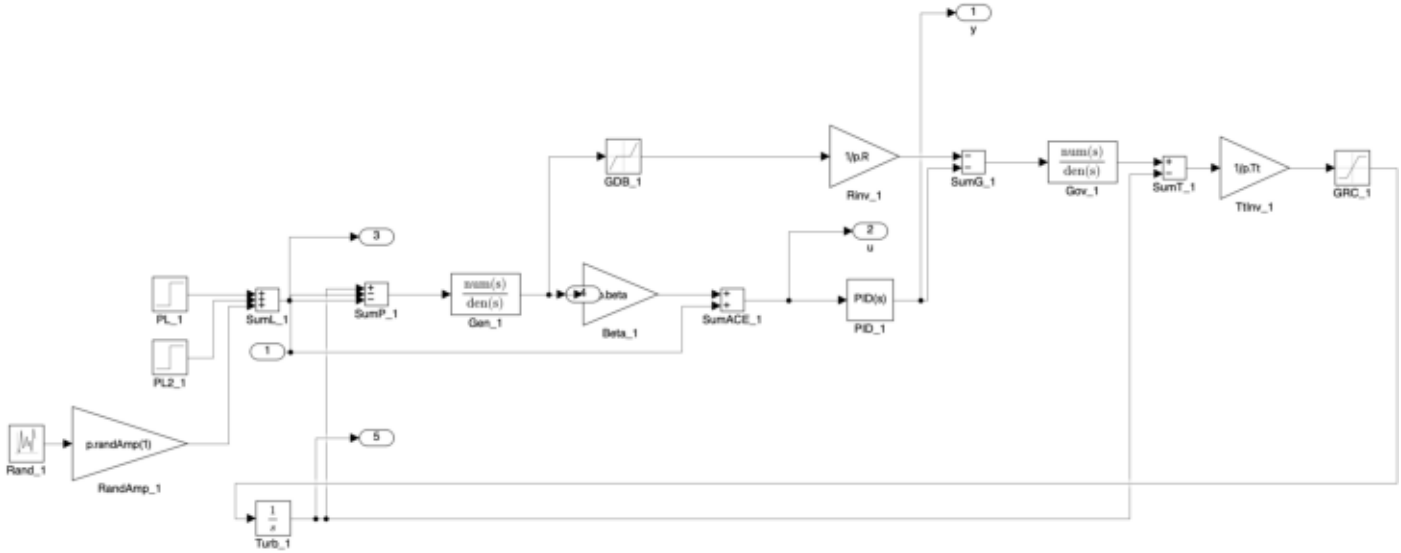
Parameters (identical areas)

Symbol	Value
Tg	0.08 s
Tt	0.3 s
Tp	20 s
Kp	120 Hz/pu
R	2.4 Hz/pu

Symbol	Value
2π·T12	0.545
β = 1/R + 1/Kp	0.425
f0	50 Hz
PID filter N	100 rad/s
Step, T	0.01 s, 100 s

Models

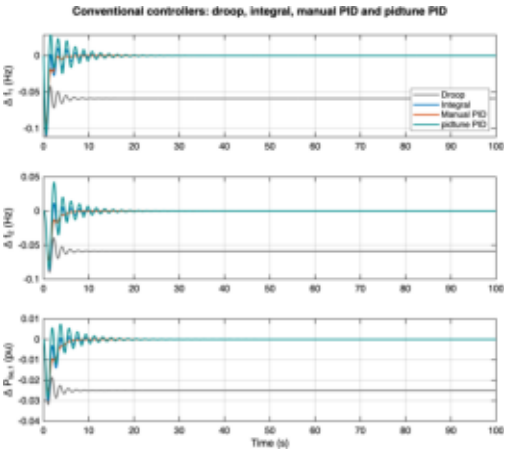
- Linear: 7-state state-space (MATLAB, Control System Toolbox)
- Simulink: fixed-step ode4, 0.01 s, generated by script
- Controller: one PID per area acting on
  ACE = ΔPtie + βΔf
- Simulink vs linear: max Δf difference 1e-7 Hz



Simulink model, Area 1 (Area 2 identical): load, power system, ACE, PID, governor, turbine

Results

Controller	KP / KI / KD	ITAE	min Δf1 (Hz)	ts Δf1 (s)	final Δf1 (Hz)	Δf1 ≥ -0.10 Hz
Droop only	0 / 0 / 0	713.25	-0.1117	8.75	-0.0588	No
Integral	0 / 0.5 / 0	0.827	-0.1089	13.5	0	No
Manual PID	0 / 0.5 / 0.1	0.482	-0.0950	6.3	0	Yes
pidtune PID	0 / 0.849 / 0	1.882	-0.1072	22.0	0	No



Findings

- Droop alone leaves Δf = -0.0588 Hz (theory -ΔPL/2β = -0.0588).
- Integral action removes the error; stable only for KI < 1.385.
- Adding D (KD = 0.1) cuts ITAE 0.827 → 0.482; adding P worsens it.
- Only the manual PID keeps the dip inside the -0.10 Hz limit.